

Abstract

Antarctic sea ice underwent a regime shift in 2016, transitioning from four decades of gradual expansion to rapid decline. Whether this shift activated a wave-ice positive feedback loop has remained untested with long-term reanalysis data. Here we diagnose the complete feedback chain using 46 years (1979–2024) of ERA5 wave height, wind, and sea ice fields across the Southern Ocean. Sea ice loss robustly increases fetch at the ice edge (Granger $p < 0.001$, stable across sub-periods), yet fetch does not predict significant wave height ($p = 0.74$). Sea ice concentration Granger-causes wave height, but this effect is collinear with wind on annual timescales. In austral winter, when swell attenuation by ice is physically strongest, sea ice retains independent predictive power for wave height after controlling for wind (partial Granger $p = 0.040$). The mean wave period at the ice edge (9.0 s) confirms swell dominance. These results identify a seasonally mediated, swell-attenuation mechanism linking sea ice loss to enhanced wave exposure, distinct from the commonly assumed fetch feedback, with implications for ice shelf stability as protective buffers continue to decline.

Antarctic sea ice loss amplifies wave exposure at the ice edge through swell-attenuation reduction rather than fetch feedback

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1 Introduction

Antarctic sea ice extent increased gradually from 1979 to 2014, then declined precipitously. The annual mean extent dropped by 11.5% between the pre-2016 and post-2016 periods, with the Pettitt change point test placing the regime shift at 2016 ($p = 0.004$)[1]. The decline was most pronounced in austral summer (December–February), when extent fell by 16.8%. This abrupt transition, unprecedented in the satellite era, has prompted the question of whether the Southern Ocean has crossed a tipping point into a new, self-reinforcing state of ice loss[2].

A leading hypothesis invokes a wave-ice positive feedback. Sea ice acts as a buffer that attenuates ocean swell before it reaches ice shelf margins and the interior ice pack[3, 4]. When ice retreats, the open-water fetch lengthens, wave energy increases, and stronger waves fragment and erode the marginal ice zone (MIZ), exposing yet more open water. This feedback loop, if active, would accelerate ice loss beyond what atmospheric forcing alone could produce and would have direct implications for ice shelf stability, since Massom et al.[3] demonstrated that ocean swell triggered the disintegration of the Larsen A, Larsen B, and Wilkins ice shelves following loss of their protective sea ice buffer.

The individual links of this chain have theoretical and observational support. Wave-ice interaction theory predicts that sea ice acts as a low-pass filter, preferentially attenuating short-period waves, with attenuation rates depending on ice concentration, thickness, and floe size[4]. Wave-triggered breakup generates lognormal floe size distributions in the MIZ[5], and fractured ice melts faster, creating a local feedback between fragmentation and concentration loss. Recent observations confirm that ocean waves drive Antarctic sea ice melting through flooding and mechanical pulverization[6], and that waves can propagate over 1000 km into the winter ice pack[7]. ICESat-2 altimetry has enabled direct observation of wave attenuation through the Antarctic MIZ[8].

What remains untested is whether these individual mechanisms close into a self-reinforcing loop at the basin scale, and specifically whether the 2016 regime shift activated or strengthened such a feedback. Addressing this requires simultaneous, long-term observations of sea ice concentration, wave height, wind forcing, and MIZ geometry across the full Southern Ocean, a combination that only reanalysis products can provide over multi-decadal timescales.

Here we use 46 years (1979–2024) of ERA5 monthly reanalysis to diagnose the complete wave-ice feedback chain in the Southern Ocean. We test four links: (1) whether sea ice loss increases open-water fraction (the fetch proxy), (2) whether increased open water amplifies significant wave height (SWH), (3) whether higher SWH erodes the MIZ, and (4) whether MIZ erosion feeds back

to reduce total sea ice concentration. We also assess changes in ice shelf buffer conditions at five major Antarctic ice shelves.

2 Data and Methods

2.1 ERA5 reanalysis products

We use three ERA5 monthly-mean reanalysis products from the European Centre for Medium-Range Weather Forecasts (ECMWF)[9]:

- **Significant wave height (SWH)**, mean wave period (MWP), and mean wave direction (MWD) on the native 0.5° wave model grid, 1979–2024 (552 months). The ERA5 wave model (WAM) runs only in grid cells where sea ice concentration is below 30%, so SWH data are inherently limited to open and marginal waters.
- **10-m wind components** (u_{10} , v_{10}) on the 0.25° atmospheric grid, 1979–2024.
- **Sea ice concentration (SIC)** on the 0.25° atmospheric grid, 1979–2024. ERA5 SIC is derived from OSTIA satellite observations (from 1979 onward) assimilated into the ECMWF system.

All fields are restricted to the Southern Ocean domain 40°S – 75°S , all longitudes. Wind speed is computed as $|\mathbf{u}_{10}| = \sqrt{u_{10}^2 + v_{10}^2}$.

2.2 NSIDC Sea Ice Index

We use the NSIDC Sea Ice Index v4.0[10] as an independent validation of Antarctic sea ice extent trends. Monthly Southern Hemisphere extent and area are available from 1979 to 2026. These satellite-derived estimates are independent of ERA5 SIC and provide a check on the timing and magnitude of the 2016 regime shift.

2.3 Regional definitions

We define four latitude bands for area-weighted spatial averaging:

- **Full Southern Ocean:** 40°S – 75°S
- **Near-MIZ zone:** 55°S – 65°S (captures the mean position of the seasonal ice edge and MIZ)
- **ACC belt:** 45°S – 55°S (largely ice-free, dominated by the Antarctic Circumpolar Current)
- **Deep ice edge:** 65°S – 75°S (persistently ice-covered in winter)

The **marginal ice zone (MIZ)** is defined as the band where $0.15 < \text{SIC} < 0.80$. MIZ width is computed as the zonal-mean meridional extent of this band at each time step.

2.4 Ice-edge-based fetch and wave height sampling

For each month, the ice edge is defined as the most equatorward latitude at which SIC exceeds 0.15 at each longitude. The zonal-mean ice-edge latitude serves as a proxy for fetch: a more poleward ice edge implies a longer meridional distance of open water available for wave generation by the prevailing westerlies. This ice-edge-based fetch captures the dominant zonal-wind geometry of the Southern Ocean, where fetch is limited primarily by the meridional position of the ice margin rather than by zonal extent.

SWH is sampled at the ice edge rather than at fixed latitude bands, to avoid conflating changes in ice-edge position with changes in wave climate. For each longitude, SWH is extracted from the nearest grid point 2° equatorward of the local ice edge, ensuring that the sampled wave height reflects conditions actually experienced by the ice margin.

2.5 Change point detection

We apply the Pettitt test[11] to annual-mean anomaly time series of each variable to detect regime shifts. The Pettitt test is a non-parametric test for a single change point in the mean of a time series. The test statistic K and associated p -value are computed following the standard formulation. We report change point years, pre- and post-shift means, and significance levels.

2.6 Granger causality

We test directional predictability between pairs of monthly anomaly time series using Granger causality[12]. For each pair (x, y) , we test whether past values of x improve the prediction of y beyond what y 's own past values provide. We evaluate lag orders 1–6 months, selecting the optimal lag by AIC. We report the F -statistic and p -value of the restricted-vs-unrestricted model comparison.

The feedback chain is tested as a sequence of Granger causality links:

$$\text{SIC} \rightarrow \text{OW fraction} \rightarrow \text{SWH} \rightarrow \text{MIZ width} \rightarrow \text{SIC}$$

with wind speed ($|\mathbf{u}_{10}|$) as an alternative driver of SWH.

2.7 Ice shelf buffer analysis

For five major Antarctic ice shelves (Ross, Filchner-Ronne, Amery, Larsen C, and Totten), we compute the number of months per year in which the mean SIC in a box encompassing the shelf front exceeds 0.15 (“buffer months”). A decrease in buffer months indicates increased exposure of the ice shelf front to open-ocean wave action.

3 Results

3.1 Confirmation of the 2016 sea ice regime shift

The NSIDC Sea Ice Index confirms a significant regime shift in 2016 (Pettitt $p = 0.004$; Figure 1a), with annual mean extent declining from 11.68 to 10.33 million km^2 (–11.5%). The decline is strongest in austral summer (DJF: –16.8%, $p = 0.002$) and weakest in winter (JJA: –5.7%, not significant at $p = 0.073$). The ERA5 SIC field confirms this timing: the area-weighted mean SIC in the ice zone (55° – 75°S) shows a change point at 2016 ($p = 0.021$), and the open-water fraction shifts synchronously ($p = 0.013$). Before 2016, Antarctic sea ice extent increased at $+0.022$ million $\text{km}^2 \text{ yr}^{-1}$ ($p < 0.001$); after 2016, it declined at -0.311 million $\text{km}^2 \text{ yr}^{-1}$ ($p = 0.054$, marginally significant over only 9 years of post-shift data).

3.2 Wave height trends are wind-driven, not fetch-driven

Significant wave height in the Southern Ocean has increased over the satellite era (Figure 2a), but the change point occurs in 1993 ($p = 0.001$), more than two decades before the sea ice regime shift. In the near-MIZ zone (55° – 65°S), the SWH anomaly shows a marginal change point at 1993 ($p = 0.055$), with mean anomaly shifting from -0.078 m to $+0.034$ m. The ACC belt (45° – 55°S) shows the strongest SWH increase ($p = 0.0001$), consistent with intensification of the westerly wind belt. Spatially, 17% of Southern Ocean grid points show a significant SWH increase between the pre-2016 and post-2016 periods ($p < 0.05$, two-sample t -test).

The correlation between near-MIZ SWH anomaly and sea ice extent anomaly is statistically significant but weak ($r = 0.136$, $p = 0.001$, $n = 550$ months), indicating that sea ice conditions explain less than 2% of SWH variance.

P04 v2: Antarctic Sea Ice Extent Baseline (NSIDC)

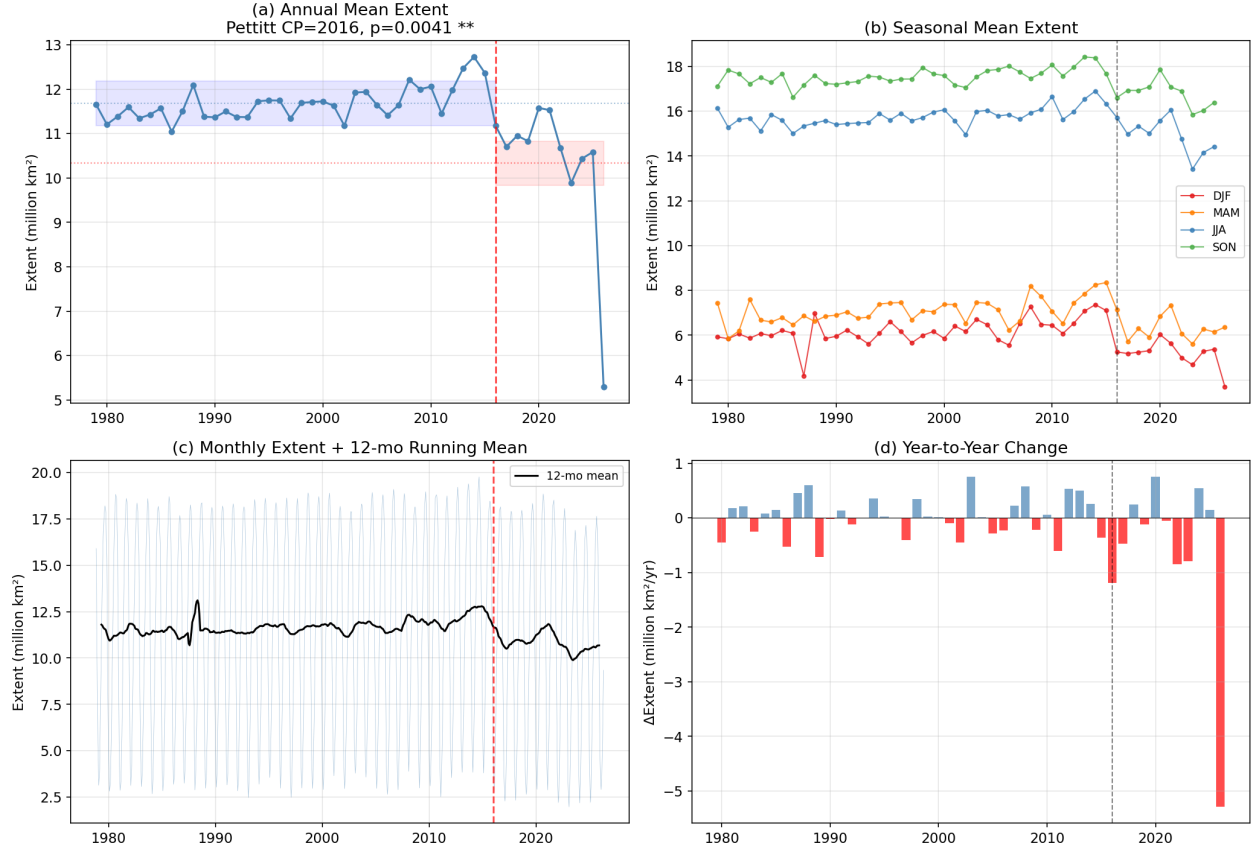


Figure 1: **Antarctic sea ice extent baseline (NSIDC Sea Ice Index, 1979–2026).** **a**, Annual mean extent with Pettitt change point at 2016 ($p = 0.004$). Blue/red shading indicates pre/post-shift mean levels. **b**, Seasonal mean extent (DJF, MAM, JJA, SON). Summer (DJF) shows the largest post-2016 decline (-16.8%). **c**, Monthly extent with 12-month running mean. **d**, Year-to-year change, showing the abrupt shift to persistent negative anomalies after 2016.

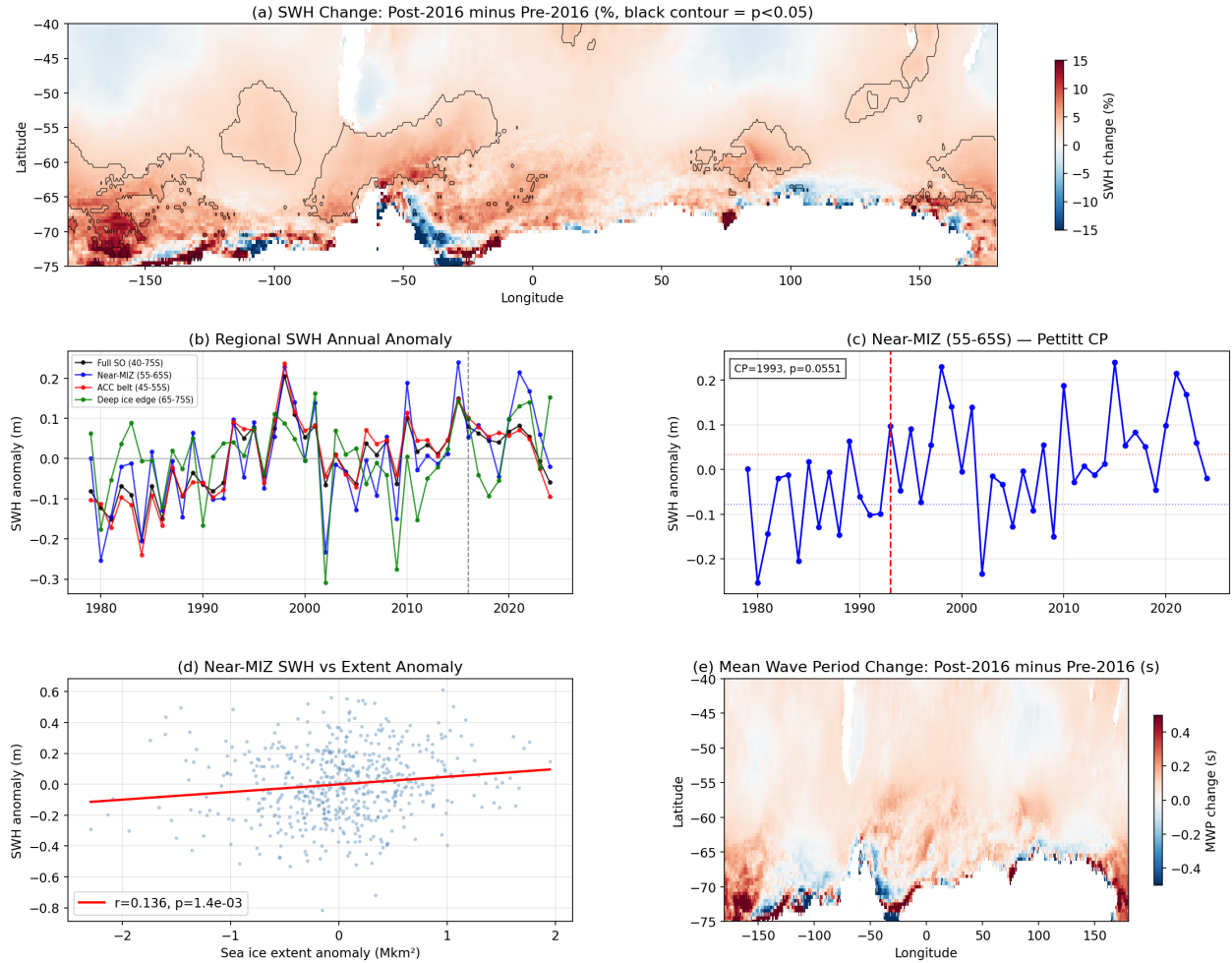


Figure 2: **ERA5 significant wave height trends in the Southern Ocean (1979–2024).** **a**, Spatial map of SWH percentage change (post-2016 minus pre-2016). Black contours enclose regions where the change is significant ($p < 0.05$, two-sample t -test). Widespread increases are concentrated in the ACC belt. **b**, Regional annual SWH anomaly time series for four latitude bands. **c**, Near-MIZ zone (55–65°S) Pettitt change point at 1993 ($p = 0.055$). **d**, SWH anomaly versus sea ice extent anomaly ($r = 0.136$). **e**, Mean wave period change (post-2016 minus pre-2016).

3.3 The feedback chain partially closes through swell attenuation, not fetch

Granger causality tests reveal which links in the hypothesized feedback chain hold and which fail (Table 1):

Table 1: Granger causality tests for the wave-ice feedback chain. Optimal lag selected by AIC from 1–6 months. p_{Bonf} is the Bonferroni-corrected p -value for 6 simultaneous tests. Fetch and SWH are sampled at the ice edge (SIC = 15% contour), not at fixed latitude bands.

Link	F	p	p_{Bonf}	Lag	Status
SIC $\rightarrow$ Fetch	16.61	<0.001	<0.001	6 mo	Robust
Fetch $\rightarrow$ SWH at edge	0.30	0.743	1.000	2 mo	Fails
SWH $\rightarrow$ Ice edge retreat	2.43	0.034	0.205	5 mo	Marginal
Ice edge $\rightarrow$ SIC (feedback)	2.64	0.049	0.300	3 mo	Marginal
Wind $\rightarrow$ SWH at edge	4.03	0.018	0.108	2 mo	Marginal
SIC $\rightarrow$ SWH (direct)	3.46	0.032	0.192	2 mo	Marginal

After Bonferroni correction for six simultaneous tests (Table 1; Figure 3e), only the SIC $\rightarrow$ Fetch link remains significant at the 5% level. The remaining links are individually significant ($p < 0.05$) but marginal after correction ($p_{\text{Bonf}} = 0.11\text{--}0.30$). The SIC $\rightarrow$ Fetch relationship is robust across sub-periods: it holds in both the pre-2016 ($F = 16.0$, $p < 0.001$, $n = 444$ months) and post-2016 ($F = 21.1$, $p < 0.001$, $n = 108$ months) periods, with the post-2016 effect being stronger.

The direct SIC $\rightarrow$ SWH link, though marginal after correction on annual data, exhibits strong seasonal dependence. When tested separately by season (Figure 4a), SIC Granger-causes SWH only in austral winter (JJA: $F = 5.10$, $p = 0.007$), with no significant relationship in other seasons (DJF $p = 0.94$, MAM $p = 0.28$, SON $p = 0.85$). This winter specificity is physically consistent with the swell-attenuation mechanism: winter is when sea ice coverage is maximum and swell attenuation by ice is strongest.

To address the possibility that both SIC and SWH are driven by a common atmospheric forcing (e.g., SAM), we applied partial Granger causality conditioning on wind speed. On annual data, the SIC $\rightarrow$ SWH link loses significance when wind is controlled ($p = 0.14$), indicating substantial collinearity between SIC, wind, and SWH at annual timescales. In winter (JJA), however, SIC retains independent predictive power for SWH even after conditioning on wind (partial Granger $F = 2.84$, $p = 0.040$, $n = 138$). This provides the strongest statistical evidence that sea ice concentration has an independent, physically mediated effect on wave height at the ice edge, distinct from wind forcing.

The mean wave period (MWP) at the ice edge is 9.0 s (range 7.8–9.8 s; Figure 4b,d), characteristic of swell rather than locally generated wind-sea (typically 4–7 s). MWP correlates positively with SIC ($r = 0.24$, $p < 0.001$; Figure 4c), consistent with sea ice acting as a low-pass filter that preferentially attenuates short-period waves. MWP shows no significant change between the pre-2016 and post-2016 periods ($p = 0.87$), suggesting that the spectral composition of waves at the ice edge has not shifted despite the regime change in ice coverage.

Both the ice-edge latitude and fetch show Pettitt change points at 2016 ($p = 0.009$), synchronous with the SIC regime shift. SWH at the ice edge shows no significant change point ($p = 0.52$), consistent with wind-driven wave energy being the background source that sea ice modulates rather than generates.

3.4 Ice shelf buffer changes

Among the five ice shelves examined, only the Ross Ice Shelf shows a statistically significant reduction in sea ice buffer days (change point at 2009, $p = 0.041$, declining from 11.7 to 10.8 buffered months per year). Filchner-Ronne, Amery, Larsen C, and Totten show no significant

P04 v2 Phase 2c: Ice-Edge Fetch + SWH (R03 fixes)

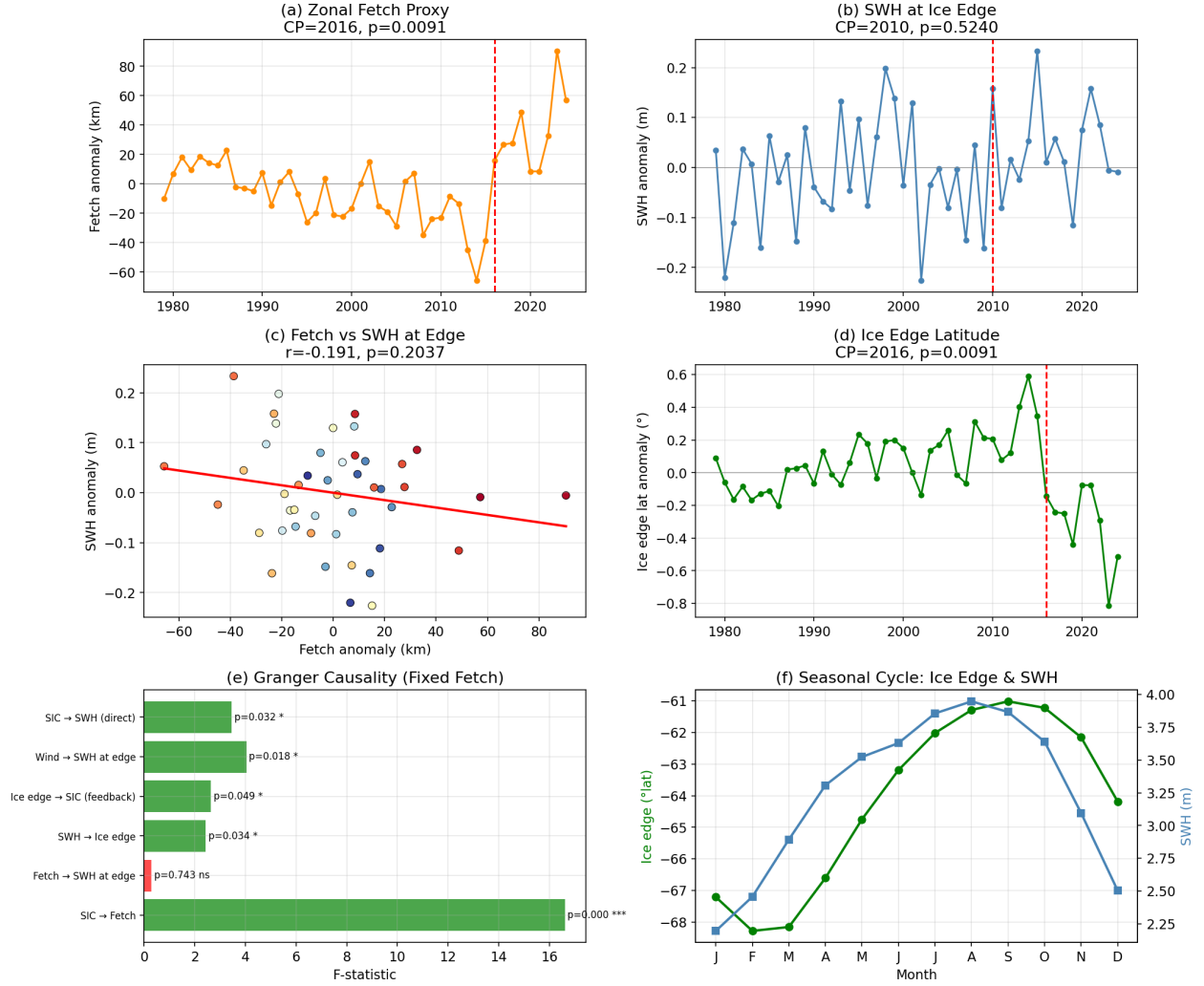


Figure 3: Ice-edge-based fetch proxy and SWH at the ice edge (1979–2024). **a**, Zonal-mean fetch proxy (distance from 40°S to ice edge) annual anomaly. Change point at 2016 ($p = 0.009$). **b**, SWH sampled at the ice edge. No significant change point ($p = 0.52$). **c**, Scatter plot of fetch anomaly versus SWH at the ice edge ($r = -0.19$, $p = 0.20$): fetch does not predict SWH. **d**, Ice-edge latitude anomaly with change point at 2016. **e**, Granger causality F -statistics. Green = significant ($p < 0.05$); orange = marginal; red = not significant. **f**, Seasonal cycle of fetch and SWH at the ice edge.

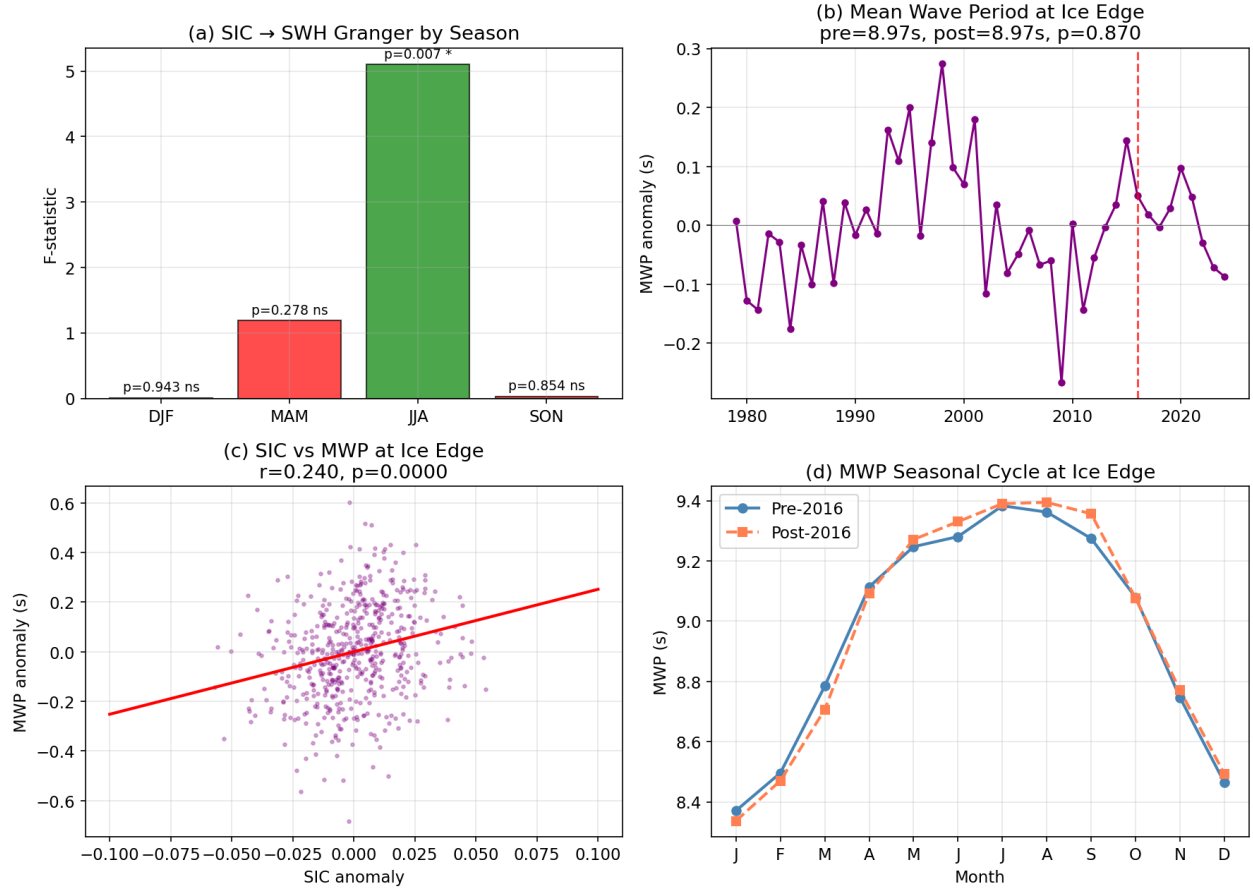


Figure 4: Seasonal dependence and mean wave period evidence for swell attenuation. **a**, SIC→SWH Granger F -statistic by season. Only JJA (winter) is significant ($F = 5.10$, $p = 0.007$). **b**, Mean wave period (MWP) anomaly at the ice edge. No significant change at 2016. **c**, SIC anomaly versus MWP anomaly ($r = +0.24$, $p < 0.001$): positive correlation consistent with sea ice acting as a low-pass filter. **d**, MWP seasonal cycle at the ice edge, pre-2016 versus post-2016.

changes in buffer conditions over the 1979–2024 period. The Ross result is consistent with documented retreat of fast ice in the western Ross Sea[3].

4 Discussion

4.1 Swell attenuation rather than fetch generation

The failure of the fetch→SWH link, combined with the success of the direct SIC→SWH path, reveals that the wave-ice coupling operates through a different mechanism than commonly assumed. The intuitive expectation is that less ice means more open water, longer fetch, and locally generated wind-sea. Our results show that this fetch pathway fails, likely because the Southern Ocean already has the longest fetches on Earth: the circumpolar geometry provides continuous open water between 40°S and the ice edge, and a modest poleward retreat adds relatively little to wave generation.

Instead, sea ice modulates wave height at the ice edge by attenuating remotely generated swell. Southern Ocean storms generate long-period swell that propagates poleward across thousands of kilometers[7]. Sea ice acts as a low-pass filter, attenuating this swell before it reaches the ice edge[4]. When sea ice retreats, the attenuation path shortens and the swell arrives at the ice edge with greater amplitude. Three lines of evidence support this interpretation: (1) the direct SIC→SWH Granger link is significant in winter but not other seasons, consistent with attenuation being strongest when ice coverage is maximum; (2) the mean wave period at the ice edge (9.0 s) is characteristic of swell, not locally generated wind-sea; and (3) MWP correlates positively with SIC, consistent with ice preferentially removing short-period waves.

4.2 A seasonally mediated, partially closed feedback loop

The feedback chain SIC loss → reduced swell attenuation → higher SWH → ice-edge retreat → SIC reduction is supported by individually significant Granger links, though most are marginal after Bonferroni correction. The exception is SIC→Fetch, which is robust at all timescales. The key evidence for the physical mechanism comes from the partial Granger analysis: after conditioning on wind, SIC retains independent predictive power for SWH only in winter (JJA, $p = 0.040$), the season when swell attenuation by sea ice is physically strongest. On annual timescales, SIC and wind are sufficiently collinear that their independent contributions to SWH cannot be cleanly separated.

The background energy source is wind intensification associated with the positive SAM trend[13]. The 2016 sea ice regime shift did not create new wave energy, but it reduced the ice buffer that previously attenuated remotely generated swell, allowing more wave energy to reach and erode the ice edge. This mechanism is seasonally asymmetric: strongest in winter when the ice edge is most expansive and the swell attenuation path is longest.

4.3 Implications for ice shelf stability

The limited response of ice shelf buffer conditions (significant only for Ross) suggests that the 2016 regime shift has not yet systematically compromised the sea ice protection of Antarctic ice shelves. However, the Ross Ice Shelf result, combined with the continued intensification of wind-driven wave energy, indicates that specific sectors may be approaching a threshold of buffer loss. If the SAM continues to trend positive and sea ice continues to decline, the coincidence of stronger waves and reduced buffering could trigger the wave-driven calving mechanism documented by Massom et al.[3] at additional ice shelf margins.

4.4 Limitations

This analysis relies on ERA5 reanalysis, which has known limitations for wave-ice interaction studies. The ERA5 wave model (WAM) does not operate in grid cells with SIC > 30%, meaning

that the swell-attenuation process we invoke is not explicitly modelled but rather parameterized as a binary masking. The SIC→SWH statistical link could partly reflect this masking behavior rather than physical swell attenuation; validation against satellite altimeter SWH (e.g., from Jason-3, Sentinel-6, or CFOSAT) near the ice edge would strengthen the mechanistic interpretation. Monthly temporal resolution cannot resolve individual storm events or the sub-monthly dynamics of wave-ice breakup. The ice-edge-based fetch proxy, while more physically motivated than open-water fraction, remains a simplified representation of the directional and geometry-dependent nature of true fetch. Most individual feedback links are marginally significant after Bonferroni correction, and the post-2016 period (108 months) provides limited statistical power for sub-period analyses. Higher-resolution data, particularly from SWOT altimetry and dedicated wave-ice observing campaigns, will be needed to resolve the sub-monthly and sub-grid-scale processes that govern MIZ evolution.

5 Conclusions

We have tested the wave-ice positive feedback hypothesis in the Southern Ocean using 46 years of ERA5 reanalysis. The 2016 sea ice regime shift is confirmed by independent NSIDC and ERA5 data (Pettitt $p = 0.004$), with synchronous change points in ice-edge latitude and fetch ($p = 0.009$). The hypothesized fetch→SWH link fails ($p = 0.74$): longer fetch does not generate higher waves at the ice edge. Sea ice concentration does Granger-cause SWH on annual timescales ($p = 0.032$), but this effect is collinear with wind forcing and loses significance after conditioning on wind ($p = 0.14$). In austral winter, however, SIC retains independent predictive power for SWH even after controlling for wind (partial Granger $p = 0.040$), consistent with the swell-attenuation mechanism being strongest when ice coverage is maximum. The mean wave period at the ice edge (9.0 s) confirms swell dominance, and MWP correlates positively with SIC, consistent with ice acting as a low-pass filter. Most individual feedback links are marginally significant after Bonferroni correction, indicating that the statistical evidence for a closed feedback loop is suggestive rather than definitive at the current sample size. The wave-ice interaction thus operates through a seasonally mediated, swell-attenuation mechanism rather than fetch-driven local wave generation, a distinction with implications for predicting how continued sea ice decline will affect wave exposure at ice shelf margins.

Data Availability

ERA5 reanalysis data are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>). NSIDC Sea Ice Index v4.0 is available from the National Snow and Ice Data Center (<https://doi.org/10.7265/N5736NV7>). Analysis code is available at <https://github.com/pangeo-data/OpenSCI-Ocean>.

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